

Experiment 3: Random Height + Fixed Wind + Fixed Obstacle | random_h_fixed_w_fixed_obs | Comparison Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	1989	11.326	13.702	21.822	15.5%	30.0%	35.0%	7.641	18.219	4.775	10.366	0.411	0.971	1.310
PPO-GRU	1992	3.128	0.723	13.852	65.0%	81.2%	82.7%	7.413	17.320	6.009	13.525	0.358	0.749	1.045
PPO-RNN	1975	2.048	1.060	3.437	46.9%	76.4%	88.1%	13.705	28.441	6.744	14.349	0.442	1.188	1.425
MPC	1003	5.962	2.910	16.131	2.5%	17.4%	53.7%	2.955	3.728	1.574	2.700	0.150	0.148	0.175

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	1989	99.5%	11	0.6%	0.6%	11	100.0%	0	0
PPO-GRU	2000	1992	99.6%	8	0.4%	0.4%	8	100.0%	0	0
PPO-RNN	2000	1975	98.8%	25	1.2%	1.2%	25	100.0%	0	0
MPC	2000	1003	50.1%	997	49.9%	49.9%	997	100.0%	0	0

Cross-Case Check: PPO/PPO-GRU/PPO-RNN on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	997	990	99.3%	647	65.4%	11.456
PPO-GRU	997	993	99.6%	176	17.7%	3.359
PPO-RNN	997	982	98.5%	120	12.2%	2.093